

# Quiz 1 — Sample 3

DATA 412/612

Write your answers in an R script. You may use R and RStudio. You may **not** use the internet or your notes.

1. Create the following matrix:

|       | [,1] | [,2] | [,3] | [,4] | [,5] | [,6] | [,7] | [,8] | [,9] | [,10] |
|-------|------|------|------|------|------|------|------|------|------|-------|
| [1,]  | 990  | 980  | 970  | 960  | 950  | 940  | 930  | 920  | 910  | 900   |
| [2,]  | 890  | 880  | 870  | 860  | 850  | 840  | 830  | 820  | 810  | 800   |
| [3,]  | 790  | 780  | 770  | 760  | 750  | 740  | 730  | 720  | 710  | 700   |
| [4,]  | 690  | 680  | 670  | 660  | 650  | 640  | 630  | 620  | 610  | 600   |
| [5,]  | 590  | 580  | 570  | 560  | 550  | 540  | 530  | 520  | 510  | 500   |
| [6,]  | 490  | 480  | 470  | 460  | 450  | 440  | 430  | 420  | 410  | 400   |
| [7,]  | 390  | 380  | 370  | 360  | 350  | 340  | 330  | 320  | 310  | 300   |
| [8,]  | 290  | 280  | 270  | 260  | 250  | 240  | 230  | 220  | 210  | 200   |
| [9,]  | 190  | 180  | 170  | 160  | 150  | 140  | 130  | 120  | 110  | 100   |
| [10,] | 90   | 80   | 70   | 60   | 50   | 40   | 30   | 20   | 10   | 0     |

2. Find the sum of the entries of the even columns of the above matrix.

3. Suppose you have the following data frame:

```
df <- data.frame(
  Name = c("Johnny", "Alisson", "Andy", "Sarah", "Alex"),
  Age = c(25, 30, 28, 22, 32),
  Annual_Income = c(50000, 60000, 55000, 48000, 70000),
  Account_Balance = c(1700, 2000, 1800, 900, 2500.90)
)
```

- What is the age average of people who get more than \$51k in a year.
  - What are their names?
4. Given a data frame `df` with only positive values, write a function that checks whether the number of columns is less than or equal to 1 or whether it has any negative values. If yes, it returns `-1`. If not, the function compares the geometric means (using the pipe operator) of the first and the second columns and produces 1 if that is correct, otherwise 0. Try your function on these inputs:

```
data.frame(a = c(9, 1, 8))
data.frame(a = c(9, 9, 8), b = c(9, 7, 6))
data.frame(a = c(9, 1, 8), b = c(9, 17, 6))
```